Companion
to

ChatGPT Teaches TikZ

Chapter 14: Clipping and Transparency (I)

Contents

14.1 What This Companion Adds

Chapter 14 introduced clipping and transparency as two ways to control what a reader sees. This Companion develops a disciplined workflow: isolate clipping inside a scope, separate fill from outline opacity, and use visual emphasis without changing the underlying geometry.

Design

Draw the mathematical object once. Use clipping to choose where it is visible and opacity to choose how strongly it is visible.

14.2 A Visibility Vocabulary

Option or command	Purpose
`\clip`	Sets the visible region for later commands.
`scope`	Prevents a clipping path from affecting unrelated objects.
`opacity=.5`	Changes fill, outline, and text opacity together.
`fill opacity=.5`	Changes only the fill opacity.
`draw opacity=.5`	Changes only the outline opacity.
`text opacity=1`	Keeps text fully visible.
`even odd rule`	Treats nested subpaths as alternating filled and empty regions.
`filldraw`	Applies a fill and an outline to one path.

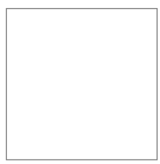
14.3 Clipping Belongs in a Scope

A clipping command affects every drawing operation that follows in the same scope.

```
\begin{tikzpicture}
  \begin{scope}
    \clip (-1,-1) rectangle (1,1);
    \draw[very thick] (0,0) circle (1.6);
  \end{scope}

  \draw[gray] (-1,-1) rectangle (1,1);
\end{tikzpicture}
```

See



The circle is complete in the source. The square is only a viewing window.

Common Mistake

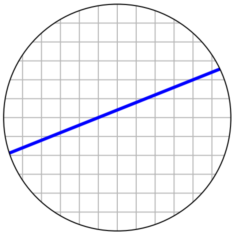
Place `\clip` before the objects it should affect. A clipping command cannot hide objects that have already been drawn.

14.4 Clipping a Grid to a Circle

The clipping path may be curved.

```
\begin{scope}
  \clip (0,0) circle (1.5);
  \draw[step=.25,gray!55]
    (-2,-2) grid (2,2);
  \draw[blue,very thick]
    (-2,-.7) -- (2,.9);
\end{scope}
\draw (0,0) circle (1.5);
```

See



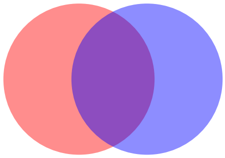
Draw the boundary after the scope when it should remain crisp and complete.

14.5 Overall Opacity

The option `opacity` changes every component of an object.

```
\fill[red,opacity=.45] (0,0) circle (1);
\fill[blue,opacity=.45] (.9,0) circle (1);
```

See



The overlap remains visible because neither disk is opaque.

14.6 Separate Fill, Outline, and Text Opacity

Overall opacity can make labels and boundaries too faint. Separate options give more control.

```
\node[
  draw=blue,
  fill=blue!30,
  fill opacity=.35,
  draw opacity=1,
  text opacity=1
] {Important label};
```

The fill recedes while the outline and text remain fully visible.

Design

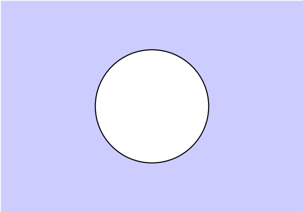
Fade the fill first; keep the outline and text clear.

14.7 Creating a Hole

The even–odd rule alternates between filled and unfilled nested regions.

```
\fill[blue!20,even odd rule]
  (-2,-1.4) rectangle (2,1.4)
  (0,0) circle (.75);
\draw (0,0) circle (.75);
```

See



Both shapes belong to one compound path. The inner circle becomes a hole.

14.8 A Spotlight Effect

A translucent compound path can dim everything except one region.

```
\fill[black,fill opacity=.18,even odd rule]
  (-2.4,-1.5) rectangle (2.4,1.5)
  (.7,.2) circle (.65);
```

The circular opening remains undimmed. This is often clearer than giving the important object a very bright color.

14.9 Worked Project: A Focused Coordinate View

The project combines a circular clipping window with one translucent region.

14.9.1 Stage 1: Establish the Window

```
\begin{scope}
  \clip (0,0) circle (1.8);
  \draw[step=.3,gray!50] (-2,-2) grid (2,2);
\end{scope}
```

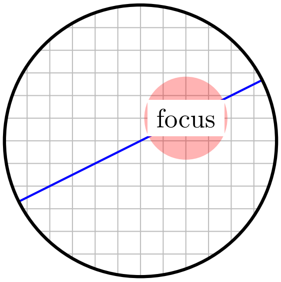
14.9.2 Stage 2: Add the Mathematical Objects

```
\draw[very thick] (0,0) circle (1.8);
\draw[blue,thick] (-2,-1) -- (2,1);
\fill[red,fill opacity=.3] (.6,.3) circle (.55);
```

14.9.3 The Complete Picture

```
\begin{tikzpicture}
  \begin{scope}
    \clip (0,0) circle (1.8);
    \draw[step=.3,gray!50] (-2,-2) grid (2,2);
    \draw[blue,thick] (-2,-1) -- (2,1);
    \fill[red,fill opacity=.3] (.6,.3) circle (.55);
  \end{scope}
  \draw[very thick] (0,0) circle (1.8);
  \node[fill=white] at (.6,.3) {focus};
\end{tikzpicture}
```

See



Design

The scope controls visibility, the line records the mathematical relation, and the translucent disk directs attention without hiding the grid.

14.10 Debugging Workshop

14.10.1 The Clip Affects Later Material

End the clipping scope before drawing objects that should be unrestricted.

14.10.2 The Boundary Disappears

Draw the boundary after the clipped material or draw it in a separate scope.

14.10.3 The Label Is Too Faint

Use `fill opacity` rather than overall `opacity`, and keep `text opacity=1`.

14.10.4 The Hole Is Filled

Place the outer and inner boundaries in the same path and add `even odd rule`.

14.11 Exercises

1. Clip a rectangular grid to a circular window.
2. Clip a large circle to a square window.
3. Compare two overlapping disks at opacity .25, .5, and .75.
4. Make a transparent fill with a fully opaque outline.

5. Keep a node label opaque while fading its fill.
6. Cut two circular holes from one rectangular region.
7. Add a spotlight opening to a dim overlay.
8. Replace an unnecessary bright highlight with controlled opacity.

14.12 Solutions

14.12.1 Exercises 1–4

```
\begin{scope}
  \clip (0,0) circle (1.5);
  \draw[step=.25] (-2,-2) grid (2,2);
\end{scope}
```

```
\begin{scope}
  \clip (-1,-1) rectangle (1,1);
  \draw (0,0) circle (1.6);
\end{scope}
```

Exercise 3 changes the same `opacity` value on both disks.

```
\filldraw[fill=blue,fill opacity=.3,draw opacity=1]
  (0,0) circle (1);
```

14.12.2 Exercises 5–7

```
\node[fill=blue,fill opacity=.3,text opacity=1]
  {label};
```

```
\fill[even odd rule]
  (-2,-1) rectangle (2,1)
  (-.7,0) circle (.35)
  (.7,0) circle (.35);
```

Exercise 7 uses the same compound-path method with a translucent outer fill. For Exercise 8, change only the visual emphasis; preserve every coordinate.

14.13 ChatGPT Workshop

Ask ChatGPT to preserve the geometry while evaluating visibility and emphasis.

1. Identify exactly which commands are affected by each clipping scope.
2. Rewrite this global clip so that later labels remain visible.
3. Separate fill, outline, and text opacity without changing the object.
4. Diagnose why this compound path has no hole.
5. Reduce decorative transparency while preserving the intended focus.

14.14 Final Checklist

Before continuing to Chapter 15, check that you can

1. place clipping inside a local scope;
2. distinguish geometry from its visible portion;
3. clip a grid or curve to a chosen window;
4. control fill, outline, and text opacity separately;
5. create a hole with the even–odd rule;
6. combine clipping and transparency without obscuring labels; and
7. use visual emphasis only when it communicates structure.